

RONALDO MENEZES

University of Exeter
Department of Computer Science
Exeter, UK

Email: r.menezes@exeter.ac.uk
Email: ronaldomenezes@gmail.com
BlueSky: [@ronaldomenezes.bsky.social](https://bsky.app/profile/ronaldomenezes.bsky.social)

- EDUCATION
- ◇ **University of York**, England, UK
DPhil in Computer Science, 2000
Thesis: *Resource Management in Open Distributed Tuple Space Systems*
Advisor: Alan Wood
 - ◇ **State University of Campinas**, Brazil
MSc in Computer Science (Thesis Option), 1995
Thesis: *A Study on Models of Parallel Computation*
Advisor: João Carlos Setubal
 - ◇ **University of Fortaleza**, Brazil
BSc in Computer Science, 1992
 - ◇ **Scholarships**
 - PhD Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil)
August 1995 – August 1999
 - MSc Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil)
January 1993 - December 1994
- RESEARCH INTERESTS
- Human Mobility, Complex Networks (aka Network Science), Social Network Analysis, Swarm Intelligence, Self-Organization, Biologically-Inspired Computing, Complex Systems.
- SELECTED RESEARCH GRANTS
- ◇ *Rapid Development of Novel Optimisation Tools for Transport and Energy Networks*
PI: Ed Keedwell
Co-PIs: Ronaldo Menezes
Status: Funded by Innovate UK, 2021-2024, £166,580
 - ◇ *Methods for the Identification of Robbery, Theft of Artworks*
PI: Douglas Ferreira
External Collaborator: Ronaldo Menezes
Status: Funded by FAPERJ-Brazil, 2019-2022, R\$ 250,000 (£35,000 approximately)
 - ◇ *Understanding Human Movement in the Healthcare Sector using AI*
PI: Jacq Christmas
Co-PIs: Ronaldo Menezes
Status: Funded by Innovate UK, 2020-2022, £166,580
 - ◇ *Hot Spot Policing Randomized Control Trial in the City of Exeter*
PI: Ronaldo Menezes
Co-PIs: Marcos Oliveira, Jo Arundale
Status: Funded by Policing Lab, 2020-2021, £29,728
 - ◇ *Identification of Human Mobility Modes using Socio-Spatio-Temporal Predictive Models*
PI: Gourab Ghoshal
Co-PIs: Ronaldo Menezes, Brooke Welles
Status: Funded by US Army Research Office, 2018-2021, US\$500,000
 - ◇ *Establishment of IANE (Northeastern Center for Artificial Intelligence), part of the IARA project. Private consortium of sponsors.*
PI: Fernando Buarque
External Collaborator: Ronaldo Menezes
Status: Funded by a Private Consortium of Industries and Government Agencies, 2020, R\$7,000,000 (£1,000,000 approximately)

- ◇ *Automated Model Build for Decarbonisation and Climate Resilience*
 PI: Ronaldo Menezes
 Co-PIs: Federico Botta, Marcos Oliveira
 Status: Funded by Innovate UK, 2021, £90,751
- ◇ *Development of App and Web Interface for Automated Anuran Recognition and Mapping*
 PI: Mark Bush
 Co-PIs: Ronaldo Menezes, Eraldo Ribeiro
 Status: Funded by NSF, 2012-2018, \$348,000
- ◇ *Understanding the Predictability Limits in Human Mobility and Social Tie Formation in the Absence of Complete Data*
 PIs: Gourab Ghoshal, Ronaldo Menezes Status: Funded by Army Research Office, 2016, \$60,000
- ◇ *InToPerformance: Influence of Network Topology to the Performance of Wireless Sensor Networks*
 PI: Angelo Brayner and Ronaldo Menezes
 Status: Funded by Funcap (Brazil), 2013-2014, \$19,384
- ◇ *Generating Networks from Government Websites*
 PI: Ronaldo Menezes
 Status: Funded by Harvard University, 2013-2013, \$13,834
- ◇ *Exploring the Use of Nature-Inspired Algorithms in Software Testing Tools*
 PI: Ronaldo Menezes
 Status: Funded by CNPq (Brazil), 2005-2007, \$110,000
- ◇ *Proactive, Selective Behavior Filtering with Data restoration: A New Technique for Runtime Neutralization of Malicious Mobile Code*
 PI: James Whittaker
 Co-PIs: Ronaldo Menezes, Cem Kaner, Michael Thomason, Jens Gregor and Gerald Marin
 Status: Funded by Office of Naval Research, 2001–2005, \$4,000,000

PUBLICATIONS BOOKS

- ◇ *Methods of Statistical Physics Applied to Seismology: A Study of Earthquakes from the Viewpoint of Complex Networks*
 D. Ferreira, A. Papa, R. Menezes. LAP LAMBERT Academic Publishing, 2016.

BOOKS EDITED

- ◇ *Complex Networks XVI*
 M Macedo, A Cardillo, W Franco, A Brayner, R Menezes. Springer Nature. 2025
- ◇ *Complex Networks XV*
 F Botta, M Macedo, H Barbosa, R Menezes. Springer Nature. 2024
- ◇ *Complex Networks XIV*
 A Teixeira, F Botta, J F F Mendes, R Menezes, G Mangioni. Springer Nature. 2023
- ◇ *Complex Networks XIII*
 D Pacheco, A Teixeira, H Barbosa, R Menezes, G Mangioni. Springer Nature. 2022
- ◇ *Complex Networks XII*
 A Teixeira, D Pacheco, M Oliveira, H Barbosa, B Gonçalves, R Menezes. Springer Nature. 2021
- ◇ *Complex Networks XI*
 H Barbosa, J Gomez-Gardenes, B Gonçalves, G Mangioni, R Menezes, M Oliveira. Springer Nature. 2020
- ◇ *Proceedings of the Intelligent Data Engineering and Automated Learning–IDEAL 2019: 20th International Conference*
 H Yin, D Camacho, P Tino, AJ Tallón-Ballesteros, R Menezes, R Allmendinger. Manchester, UK, November 14–16, 2019, Springer Nature
- ◇ *Complex Networks VIII*
 B Gonçalves, R Menezes, R Sinatra, V Zlatić Springer International Publishing. 2017

- ◇ *Complex Networks VII*
H Cherifi, B Gonçalves, R Menezes, R Sinatra. Series: Studies in Computational Intelligence, Vol. 644, 2016.
- ◇ *Complex Networks V*
P Contucci, R Menezes, A Omicini, J Poncela-Casasnovas. Series: Studies in Computational Intelligence, Vol. 549, 2014.
- ◇ *Complex Networks IV*
R Menezes, A Evsukoff and M C González. Series: Studies in Computational Intelligence, Vol. 424, 2013.
- ◇ *Complex Networks: Second International Workshop, CompleNet 2010, Rio de Janeiro, Brazil, October 13-15, 2010, Revised Selected Papers*
L F Costa, A Evsukoff, G Mangioni and R Menezes. Series: Communications in Computer and Information Science, Vol. 116, 2011.
- ◇ *Complex Networks: Results of the 1st International Workshop on Complex Networks (CompleNet 2009)*
R Menezes, S Fortunato, G Mangioni and V Nicosia. Series: Studies in Computational Intelligence, Vol. 207, 2009.

BOOK CHAPTERS

- ◇ *Towards intelligent dynamics of an active transport system for biking*
Andrés Medaglia, Maria Wilches-Mogollon, Olga Sarmiento, Felipe Montes, Luis Guzman, Mauricio Sanchez-Silva, Ronaldo Menezes, Daró Hidalgo, Karla Parra, Andrés Useche, Hansel Ochoa-Montero, Natalia Rodríguez, Lorena Salamanca. In Safer Complex Systems Case Studies. Royal Academy of Engineering. May 2022.
- ◇ *Networks in Rural Sociology*
Ronaldo Menezes and Pramod Divakarmurthy. In Encyclopedia on Social Network Analysis and Mining. Reda Alhajj and Jon Rokne (Editors). August 2014.
- ◇ *Simulating Crime Against Properties Using Swarm Intelligence and Social Networks*
Vasco Furtado, Adriano Melo, André Coelho, Ronaldo Menezes and Mairon Belchior
Artificial Crime Analysis Systems: Using Computer Simulations and Geographic Information Systems. IGI Global, 2008. 300-318.
- ◇ *Scalability in Linda-like Coordination Systems*
Ronaldo Menezes, Robert Tolksdorf and Alan Wood
in Coordination of Internet Agents: Models, Technologies and Applications
A. Omicini, F. Zambonelli, M. Klush and R. Tolksdorf (editors), Springer-Verlag, 2001

UNDER REVIEW AND ONGOING

- ◇ *Systematic comparison of gender inequality in scientific rankings across disciplines* AM Jaramillo, M Macedo, M Oliveira, F Karimi, R Menezes arXiv preprint arXiv:2501.13061 (under review Nature Human Behaviour)
- ◇ *The parenthood effect in urban mobility* M Macedo, R Menezes, A Cardillo arXiv preprint arXiv:2501.02299 (under review Nature Human Behaviour)
- ◇ *A Data-Driven Supervised Machine Learning Approach to Estimating Global Ambient Air Pollution Concentrations With Associated Prediction Intervals* LJ Berrisford, H Barbosa, R Menezes arXiv preprint arXiv:2402.10248 (under review Royal Society Open Science)
- ◇ *Super-Linear Growth and Rising Inequality in Online Social Communities: Insights from Reddit* G Machado, D Pacheco, R Menezes, G Baxter arXiv preprint arXiv:2503.02661 (under review EPJ Data Science)
- ◇ *Validating Urban Scaling Laws through Mobile Phone Data: A Continental-Scale Analysis of Brazil's Largest Cities* RS Alencar, FL Ribeiro, H Samaniego, R Menezes, AG Evsukoff arXiv preprint arXiv:2503.00550 (under review Computational Urban Science)

- ◇ *Vector fields as a framework for modelling the mobility of commodities* S Farokhnejad, AS da Mata, M Macedo, R Menezes arXiv preprint arXiv:2506.02047 (under review PLoS One)
JOURNAL
- ◇ *Geographical Isolation as a Driver of Political Violence in African Cities* R Prieto-Curiel, R Menezes Nature Communications volume 16, Article number: 10049 (2025)
- ◇ *Beyond proximity: A review and framework to further understanding of greenspace accessibility in the x-minute city* A Schendl, JM Bullock, R Menezes, S Willcock People and Nature. 2025
- ◇ *A data-driven supervised machine learning approach to estimating global ambient air pollution concentrations with associated prediction intervals* LJ Berrisford, H Barbosa, R Menezes Royal Society Open Science 12 (7), 241288. 2025
- ◇ *The impact of the social exposome in cardiovascular health and disease* K El Mekkaoui, C Mukherjee, C Santana, H Silva, A Quazi, R Menezes The Journal of Precision Medicine: Health and Disease. 2025
- ◇ *A Framework for Scalable Ambient Air Pollution Concentration estimation* LJ Berrisford, LS Neal, HJ Buttery, BR Evans, R Menezes Environmental Data Science. Volume 4, e17. 2025
- ◇ *Mobility and transit segregation in urban spaces* N Iyer, R Menezes, H Barbosa Environment and Planning B: Urban Analytics and City Science 51 (7), 1496-1512. 2024
- ◇ *The role of transport systems in housing insecurity: a mobility-based analysis* N Iyer, R Menezes, H Barbosa EPJ Data Science 13 (1), 49. 2024
- ◇ *Dynamic predictability and activity-location contexts in human mobility* B Poudyal, D Pacheco, M Oliveira, Z Chen, HS Barbosa, R Menezes, G Ghoshal Royal Society Open Science 11 (9), 240115. 2024
- ◇ *Dynamics of consumption of child sexual abuse material on the dark web*
P Divakarmurthy, B Cunha, J Passoud, R Menezes, M Oliveira. 2024. PLOS one, 19(7), p.e0306516
- ◇ *Environmental Insights: Democratizing Access to Ambient Air Pollution Data and Predictive Analytics with an Open-Source Python Package*
LJ Berrisford, R Menezes. Environmental Modelling & Software, Volume 179, 2024, 106131.
- ◇ *Estimating ambient air pollution using structural properties of doad networks*
LJ Berrisford, E Ribeiro, R Menezes. Environment and Planning B: Urban Analytics and City Science. 2024
- ◇ *The structure of segregation in co-authorship networks and its impact on scientific production*
AM Jaramillo, HTP Williams, N Perra, R Menezes. EPJ Data Science 12 (1), 47. 2023
- ◇ *Transport and Mobility Segregation in Urban Spaces*
N Iyer, R Menezes, H Barbosa. International Journal of Population Data Science 8 (3). 2023
- ◇ *COVID-19 is linked to changes in the time-space dimension of human mobility*
C Santana, F Botta, H Barbosa, F Privitera, R Menezes, R Di Clemente. Nature Human Behaviour, 1-11. 2023
- ◇ *Contrasting social and non-social sources of predictability in human mobility*
Z Chen, S Kelty, A Evsukoff, BF Welles, JP Bagrow, R Menezes, G Ghoshal. Nature Communications. Volume 13, Article number: 1922. 2022.
- ◇ *Beyond exploitation: measuring the impact of local search in swarm-based memetic algorithms through the interactions of individuals in the population*
C Santana, M Oliveira, C Bastos-Filho, R Menezes. Swarm and Evolutionary Computation, 101040. 2022
- ◇ *Differences in the spatial landscape of urban mobility: gender and socioeconomic perspectives*
M Macedo, L Lotero, A Cardillo, R Menezes, H Barbosa. March 2022. PLoS One
<https://doi.org/10.1371/journal.pone.0260874>
- ◇ *Reaching to the Top: The Gender Effect in Highly-ranked Academics in Computer Science*
A Jaramillo, M Macedo, R Menezes. Advances in Complex Systems. Vol. 24, No. 03n04, 2150008. 2021.

- ◇ *Breast cancer diagnosis using thermal image analysis: A data-driven approach based on swarm intelligence and supervised learning for optimized feature selection.*
M Macedo, M Santana, WP dos Santos, R Menezes, C Bastos-Filho. *Applied Soft Computing*, 107533. 2021
- ◇ *Uncovering the social interaction network in swarm intelligence algorithms*
M Oliveira, D Pinheiro, M Macedo, C Bastos-Filho, R Menezes. *Applied Network Science* 5, 1-20. 2020
- ◇ *Uncovering the differences and similarities between physical and virtual mobility*
S Hazarie, H Barbosa, A Frank, R Menezes, G Ghoshal. *Journal of the Royal Society Interface* 17 (168), 20200250. 2020
- ◇ *Analysis of human mobility in the UK during the COVID-19 pandemic*
C Santana, F Botta, H Barbosa, F Privitera, R Menezes, R Di Clemente. preprint at github10. 2020
- ◇ *A Data-Driven Social Network Intervention for Improving Organ Donation Awareness Among Minorities: Analysis and Optimization of a Cross-Sectional Study*
MD Murphy, D Pinheiro, R Iyengar, G Lim, R Menezes, M Cadeiras. *Journal of Medical Internet Research* 22 (1), e14605. 2020
- ◇ *A data-driven network approach for characterization of political parties' ideology dynamics*
J Faustino, H Barbosa, E Ribeiro, R Menezes. *Applied Network Science* 4 (1), 48. 2019
- ◇ *An approach for thematic relevance analysis applied to textual contributions in discussion forums*
CJR Machado, AMA Maciel, RL Rodrigues, R Menezes. *International Journal of Distance Education Technologies (IJDET)* 17 (3), 37-51. 2019
- ◇ *A Data-Driven Framework for Monitoring and Improving Organ Donation Awareness among Minorities Using Optimal Social Network Intervention* D Pinheiro, M Murphy, R Iyengar, J Lazareff, J Choi, R Menezes, M Cadeiras. *The Journal of Heart and Lung Transplantation* 38 (4), S27. 2019
- ◇ *Unveiling the Interplay between the TLR4/MD2 Complex and HSP70 in the Human Cardiovascular System: A Computational Approach* AA de Oliveira, J Faustino, ME de Lima, R Menezes, KP Nunes. *International journal of molecular sciences* 20 (13), 3121. 2019
- ◇ *Spatio-temporal variations in the urban rhythm: the travelling waves of crime*
M Oliveira, E Ribeiro, C Bastos-Filho, R Menezes. *EPJ Data Science* 7 (1), 29. 2018
- ◇ *Towards evidence of long-range correlations in shallow seismic activities*
D Ferreira, J Ribeiro, A Papa, R Menezes. *EPL (Europhysics Letters)* 121 (5), 58003. 2018
- ◇ *Human mobility: Models and applications*
Hugo Barbosa-Filho, Marc Barthelémy, Gourab Ghoshal, Charlotte R James, Maxime Lenormand, Thomas Louail, Ronaldo Menezes, José J Ramasco, Filippo Simini, Marcello Tomasini. *Physics Reports*. 2018
- ◇ *The scaling of crime concentration in cities*
M. Oliveira, C. Bastos-Filho, R. Menezes. *PloS one* 12 (8), e0183110. 2017
- ◇ *On the effect of human mobility to the design of metropolitan mobile opportunistic networks of sensors*
M. Tomasini, B. Mahmood, F. Zambonelli, A. Brayner, R. Menezes. *Pervasive and Mobile Computing* 38, 215-232. 2017.
- ◇ *The Role of Human Relations and Interactions in Designing Memory-Related Models for Sensor Networks*
B. Mahmood, R. Menezes. *Sensors & Transducers* 199 (4), 42. 2016
- ◇ *The Effect of Recency to Human Mobility*
Hugo Barbosa, Fernando B Lima-Neto, Alexandre Evsukoff, and Ronaldo Menezes. *EPJ Data Science*. *EPJ Data Science* 4:21. 2015.
- ◇ *Social-Driven Information Dissemination for Mobile Wireless Sensor Networks*
Basim Mahmood, Marcello Tomasini, and Ronaldo Menezes. *Sensors and Transducers Journal*. Volume 189. Issue 6. June 2015
- ◇ *On the agreement between small-world-like OFC model and real earthquakes*
Douglas S. R. Ferreira, Andrés Papa and Ronaldo Menezes. *Physics Letters A*. Volume 379. Issue 7. March 2015

- ◇ *Using Network Science to Assess Particle Swarm Optimizers*
Marcos Oliveira, Carmelo J. A. Bastos-Filho, Ronaldo Menezes. *Social Network Analysis and Mining*. 5:3. January 2015
- ◇ *Using Social Media as a Method of Increasing Organ Donation*
Martin Cadeiras, Mohammad Soroya, Sagar Patel, Charlotte Starling, Gene Lim, Kevin Dong, Ronaldo Menezes, Mario Deng. *Circulation*. Suppl 2. November 2014.
- ◇ *Small World Picture of Worldwide Seismic Events*
Douglas S. R. Ferreira, Andrés Papa and Ronaldo Menezes. *Physica A*. Volume 408. pp. 170–180. August 2014
- ◇ *A Self-Organized Approach for Detecting Communities in Networks*
Ben Collingsworth and Ronaldo Menezes. *Social Netw. Analys. Mining*. 4:169. February 2014
- ◇ *Understanding the Spread of Malicious Mobile-phone Programs and their Damage Potential*
Pu Wang, Marta C. González, Ronaldo Menezes, Albert-László Barabási. *Int. J. Inf. Sec.* 12(5): 383-392 (2013)
- ◇ *The United States Heart and Lung Transplant "Social" Network. Insights from the UNOS Database*
S. Venugopal, E. Stoner, M. Deng, R. Menezes, M. Cadeiras. *The Journal of Heart and Lung Transplantation* (Vol. 32, Issue 4, Supplement), Pages S159-S160, 2013.
- ◇ *Understanding organ transplantation in the USA using geographical social networks*
Srividhya Venugopal, Evan Stoner, Martin Cadeiras, Ronaldo Menezes. *Social Netw. Analys. Mining* 3(3): 457-473. 2013.
- ◇ *A Bio-inspired Crime Simulation Model*
Vasco Furtado, Adriano Melo, André L. V. Coelho, Ronaldo Menezes, Ricardo Perrone. *Decision Support Systems* 48(1): 282-292. 2009.
- ◇ *Knowledge tuples never die, they just fade away*
L. Nixon, R. Tolksdorf, R. Menezes, and A. Wood. *Ubiquitous Computing and Communication Journal*, vol. 3, 2008.
- ◇ *An Adaptive In-network Aggregation Operator for Query Processing in Wireless Sensor Networks*
Angelo Brayner, Aretusa Lopes, Diorgens Meira, Ricardo Vasconcelos and Ronaldo Menezes *Journal of Systems and Software*. Volume 81, Issue 3, pp. 328–342. March 2008.
- ◇ *Intelligent Systems at Florida Tech*
Philip K. Chan, Ronaldo Menezes, Debasis Mitra, Eraldo Ribeiro, Marius Silaghi. *IEEE Intelligent Informatics Bulletin* 8(1): 5-6. 2007
- ◇ *Toward Adaptive Query Processing in Wireless Sensor Networks*
Angelo Brayner, Aretusa Lopes, Diorgens Meira, Ricardo Vasconcelos and Ronaldo Menezes *Signal Processing*, Volume 87, No. 12, December 2007, pp 2911 – 2933, Elsevier.
- ◇ *Case Studies for Self-Organization in Computer Science*
Marco Mamei, Ronaldo Menezes, Robert Tolksdorf and Franco Zambonelli
Journal of Systems Architecture, 2006, Volume 53, No. 8-9, pp. 443 – 460, Elsevier.
- ◇ *On Coordination and its Significance to Distributed and Multi-Agent Systems*
Sascha Ossowski and Ronaldo Menezes
Concurrency and Computation: Practice and Experience, Wiley. September 2005.
- ◇ *On the Semantics of Coordination Models for Distributed Systems: The LogOp Case Study*
Ronaldo Menezes, Andrea Omicini and Mirko Viroli
Electronic Notes in Theoretical Computer Science, Volume 97, pp. 97 – 124, July 2004, Elsevier
- ◇ *Resource Management in Open Linda Systems*
Ronaldo Menezes
Concurrency and Computation: Practice and Experience, Vol. 15, pp. 1233 – 1256, 2003, Wiley
- ◇ *Coordination in a Content-Addressable Web*
Ronaldo Menezes, Iain Merrick and Alan Wood
Autonomous Agents and Multi-Agent Systems Journal, Vol. 2, No. 3, pp. 287 – 301, 1999, Kluwer

citembullet *Network-Centric Analysis of Memetic Operators and Communication Topologies for Swarm Intelligence Algorithms* C Silva, C Santana, C Bastos-Filho, R Menezes. International Conference on Complex Networks, 27–40

- ◇ *Understanding the Structure and Resilience of the Brazilian Federal Road Network Through Network Science* J Taveira, FBL Neto, R Menezes International Conference on Complex Networks, 107–121
- ◇ *The Rephrased Reality: Analysing Sentiment Shifts in LLM-Rephrased Text* FL Richardson, A Villavicencio, R Menezes Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing, 920–927. 2025
- ◇ *Academic Networks and Resilience: An Education Perspective* P Velasquez-Ferrufino, R Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey, 2024.
- ◇ *Network-based Analysis of Crossover Operators in Genetic Algorithms* R Rosenbaum, R Menezes, C Santana. Complex Networks & Their Applications XIII. Istanbul, Turkey, 2024.
- ◇ *Branching Out: How Academic Genealogy Networks Shape Academic Groups* G Moraes, A Brayner, R Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey, 2024.
- ◇ *Multidimensional Analysis of the Influence of Socioeconomic and Political Indicators on the Spread of COVID-19: A Case Study of Brazilian Cities (2020-2024)* R Reis, A Brayner, M Ângelo, R Menezes. Simpósio Brasileiro de Banco de Dados (SBBD), 394-405. 2024
- ◇ *Public Transit Inequality in the Context of the Built Environment* N Iyer, R Menezes, H Barbosa. International Conference on Complex Networks, 203-218. 2024
- ◇ *Social Interactions Matter: Is Grey Wolf Optimizer a Particle Swarm Optimization Variation?* RC Lira, M Macedo, HV Siqueira, R Menezes, C Bastos-Filho. International Conference on Complex Networks, 101-114. 2024
- ◇ *Nature vs. Nurture in Science: The Effect of Researchers Segregation on Papers' Citation Histories* AM Jaramillo, F Montes, R Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet). pp. 141-154. 2023
- ◇ *Network Entropy as a Measure of Socioeconomic Segregation in Residential and Employment Landscapes* N Iyer, R Menezes, H Barbosa. Proceedings of the 14th Conference on Complex Networks (CompleNet). pp. 26-37. 2023
- ◇ *Academic Mobility as a Driver of Productivity: A Gender-centric Approach* M Macedo, AM Jaramillo, R Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet). pp. 120-131. 2023
- ◇ *Using Vector Fields in the Modelling of Movements as Flows* S Farokhnejad, M Oliveira, E Ribeiro, R Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet). pp. 155-167. 2023
- ◇ *A Data-Driven Approach to Cattle Epidemic Modelling Under Uncertainty* S Farokhnejad, D Cardoso, C Rocha, AS da Mata, R Menezes. Proceedings of the 13th Conference on Complex Networks (CompleNet). 2022
- ◇ *Networks of evolution: Modelling and deconstructing genetic algorithms using dynamic networks* Clodomir Santana, Edward Keedwell, Ronaldo Menezes. Proceedings of the Genetic and Evolutionary Computation Conference. 2022
- ◇ *Modelling the Social Interactions in Grey Wolf Optimizer* RC Lira, M Macedo, HV Siqueira, R Menezes, C Bastos-Filho 2021 IEEE Latin American Conference on Computational Intelligence (LA-CCI), 1-6. 2021
- ◇ *Fishing for interactions: a network science approach to modeling fish school search* M Macedo, L Taw, N Gurrupadi, RC Lira, D Pinheiro, M Oliveira, R Menezes. Proceedings of the Genetic and Evolutionary Computation Conference, 40-48. 2021
- ◇ *An approach to assess swarm intelligence algorithms based on complex networks* C Santana, E Keedwell, R Menezes. Proceedings of the 2020 Genetic and Evolutionary Computation Conference, 31-39. 2020

- ◇ *A Longitudinal Analysis of Vocabulary Changes in Social Media*
H Hamoodat, F Aswad, E Ribeiro, R Menezes. Complex Networks XI, 212-221. 2020
- ◇ *Communities of Human Migration in Social Media: An Experiment in Social Sensing*
F Aswad, H Hamoodat, E Ribeiro, R Menezes. Complex Networks XI, 222-232. 2020
- ◇ *Characterizing the Dynamics of Academic Affiliations: A Network Science Approach*
J Faustino, N Iyer, J Mendonza, R Menezes. Complex Networks XI, 393-404. 2020
- ◇ *Network-Based Delineation of Health Service Areas: A Comparative Analysis of Community Detection Algorithms*
D Pinheiro, R Hartman, E Romero, R Menezes, M Cadeiras. Complex Networks XI, 359-370. 2020
- ◇ *Gender Patterns of Human Mobility in Colombia: Reexamining Ravenstein's Laws of Migration*
M Macedo, L Lotero, A Cardillo, H Barbosa, R Menezes. Complex Networks XI, 269-281. 2020
- ◇ *Short Text Tagging Using Nested Stochastic Block Model: A Yelp Case Study*
J Bowllan, K Cozart, SMM Seyednezhad, A Smith, R Menezes. International Conference on Complex Networks and Their Applications, 822-833. 2019
- ◇ *Modelling the Social Interactions in Ant Colony Optimization*
N Gurrupadi, L Taw, M Macedo, M Oliveira, D Pinheiro, C Bastos-Filho, R Menezes. International Conference on Intelligent Data Engineering and Automated Learning. 216-222, Springer. 2019.
- ◇ *Characterizing the Social Interactions in the Artificial Bee Colony Algorithm* L Taw, N Gurrupadi, M Macedo, M Oliveira, D Pinheiro, C Bastos-Filho, R Menezes. 2019 IEEE Congress on Evolutionary Computation (CEC), 1243-1250. 2019
- ◇ *Social Media Vocabulary Reveals Education Attainment of Populations*
H Hamoodat, E Ribeiro, R Menezes. International Workshop on Complex Networks, 157-168. 2019
- ◇ *Entropy in network community as an indicator of language structure in emoji usage: a Twitter study across various thematic datasets*
R Hartman, SMM Seyednezhad, D Pinheiro, J Faustino, R Menezes. International Conference on Complex Networks and Their Applications, 328-337. 2018
- ◇ *Author attribution using network motifs*
Y Al Rozz, R Menezes. International Workshop on Complex Networks, 199-207. 2018
- ◇ *Complex Networks Reveal a Glottochronological Classification of Natural Languages*
H Hamoodat, Y Al-Rozz, R Menezes. International Workshop on Complex Networks, 209-219. 2018
- ◇ *Análise de Relevância Temática de Postagens em Fóruns de Discussão em Relação ao uso de Vídeos como Recurso Didático (in Portuguese)*
C Machado, A Maciel, R Rodrigues, R Menezes. Brazilian Symposium on Computers in Education, 2018
- ◇ *Improved Multi-Objective Binary Fish School for Feature Selection*
M Macedo, C. Bastos-Filho, R. Menezes. Flairs, 2018.
- ◇ *Refugee and Immigration: Twitter as a Proxy for Reality*
Firas Aswad, Ronaldo Menezes. Flairs, 2018.
- ◇ *Characterization of Users by Using Hourly and Daily Spatio-temporal Patterns Extracted from GPS Trajectories*
Marcello Tomasini, C. Bastos-Filho, Ronaldo Menezes. Flairs, 2018.
- ◇ *Emoji-word Network Analysis: Sentiments and Semantics*
Seyed Mohammad Mahdi Seyednezhad, Halley Fede, Isaiah Herrera and Ronaldo Menezes. Flairs, 2018.
- ◇ *On the Performance of Network Science Metrics as Long-Term Investment Strategies in Stock Markets*
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PEER-REVIEWED PRESENTATIONS (WITHOUT PUBLICATIONS)

- ◇ *Learning Complex Network Structures through Graph-Augmented Normalizing Flows for Enhanced Anomaly Detection in Multivariate Time Series* C Silva, AM Ribeiro, R Monteiro, R Menezes, C Bastos-Filho, D Pinheiro. XVI International Conference on Complex Networks. 2025
- ◇ *Revealing Swarm Intelligence Patterns using Network Science* RCL Silva, M Macedo, H Siqueira, R Menezes, C Bastos-Filho. XVI International Conference on Complex Networks. 2025
- ◇ *Learning Complex Network Structures through Graph-Augmented Normalizing Flows for Enhanced Anomaly Detection in Multivariate Time Series.* C Silva, A M Ribeiro, R Monteiro, R Menezes, C Bastos-Filho, D Pinheiro. XVI International Conference on Complex Networks. 2025
- ◇ *Statistics on Reddit: entropy and Gini coefficient*
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- ◇ *Academic Mobility as a Driver of Productivity: A Gender-centric Approach*
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- ◇ *Gender representation and homophily as barriers to women's career in science*
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G. Ghoshal, G. Mangioni, R. Menezes, J. Poncela-Casanovas. Social Network Analysis and Mining 4 (1), 2014
- ◇ *Special Issue on Self-Organizing Coordination*
Matteo Casadei, Marco Mamei and Ronaldo Menezes. Science of Computer Programming, Volume 78. 2013
- ◇ *Special Issue on intelligent distributed computing*
Costin Badica and Ronaldo Menezes. Concurrency and Computation: Practice and Experience 24(6). 2012
- ◇ *Special Issue on Semantic Extensions to Middleware*
Reto Krümmenacher, Elena Paslaru Bontas Simperl and Ronaldo Menezes. Journal of Software 4(8). 2009.
- ◇ *Special Issue: Coordination Models and Systems*
Sascha Ossowski and Ronaldo Menezes. Concurrency and Computation: Practice and Experience 18(4). 2006.
- ◇ *Special issue on nature-inspired applications and systems*
Muddassar Farooq and Ronaldo Menezes. Journal of Systems Architecture 52(8-9). 2006.
- ◇ *Special Issue: Computer Security*
Giampaolo Bella, Ronaldo Menezes Concurrency - Practice and Experience 16(11). 2004

TECHNICAL REPORTS

- ◇ *Towards Evidences of Long-Range Correlations in Seismic Activity*
Douglas S. R. Ferreira, Andrés Papa and Ronaldo Menezes. arXiv:1405.0307. 2014
- ◇ *New generation of mobile phone viruses and corresponding countermeasures*
Pu Wang, Marta C. González, Ronaldo Menezes, Albert-László Barabási. The Computing Research Repository. 2010.
- ◇ *On the Implementation of SwarmLinda: A Linda System based on Swarm Intelligence*
Ahmed Charles, Ronaldo Menezes and Robert Tolksdorf
Florida Institute of Technology, Department of Computer Sciences, CS-2004-03, 2004
- ◇ *A New Approach to Scalable Linda-systems Based on Swarms (Extended Version)*
Ronaldo Menezes and Robert Tolksdorf
Florida Institute of Technology, Department of Computer Sciences, CS-2003-04, 2003
- ◇ *On the Use of Linda as a Framework for Distributed Database Systems*
Madhan Thirukonda and Ronaldo Menezes
Florida Institute of Technology, Department of Computer Sciences, CS-2002-02, 2002
- ◇ *Ligia: A Java based Linda-like Run-time System with Garbage Collection of Tuple Spaces*
Ronaldo Menezes and Alan Wood
University of York, Department of Computer Science, YCS 304, 1998
- ◇ *Parallel Computing Models and Algorithm Design (In Portuguese)*
Ronaldo Menezes and João Carlos Setubal
Institute of Computing, State University of Campinas, Brazil, 1995

EXHIBITIONS ◇ **The Art of Networks II**, Poland, January 2016.

- ◇ **The Art of Networks II**, Foosaner Art Museum. September-November 2015.
- ◇ **The Art of Networks II**, New York Hall of Science. March-May 2015.
- ◇ **The Art of Networks**, Foosaner Art Museum. March-April 2012.
- ◇ **Analyse This: Digital Models of Swarm Intelligence**, Ruth Funk Center for Textile Arts. September 2010.

WORK

EXPERIENCE

ACADEMIC

- ◇ **Computer Science Director of Global Development**, University of Exeter, UK
September 2024 - Present
- ◇ **Professor of Data and Network Science**, University of Exeter, UK
August 2018 – Present
- ◇ **External Examiner**, University of Sussex, UK
January 2025 - Present
- ◇ **Visiting Professor**, Federal University of Cear , Brazil
October 2022 - 2024
- ◇ **External Examiner**, University of Warwick, UK
January 2021 - 2024
- ◇ **Alan Turing Fellow**, The Alan Turing Institute, UK
October 2021 - January 2024
- ◇ **Head of the Computer Science Department**, University of Exeter, UK
August 2018 – August 2022
- ◇ **Professor in Computer Science**, Florida Institute of Technology, USA
March 2016 – 2018
Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Computing Careers and Disciplines. Joint appointments: School of Human-Centered Design, Innovation & Art
- ◇ **Director of the School of Computing**, Florida Institute of Technology, USA
December 2016 – October 2017
- ◇ **Department Head of Computer Science**, Florida Institute of Technology, USA
December 2015 – December 2016
- ◇ **Associate Professor in Computer Science**, Florida Institute of Technology, USA
August 2006 – 2016
Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Swarm Intelligence, Introduction to Biologically Inspired Computing.
- ◇ **Visiting Researcher**, Department of Physics, Northeastern University, CCNR (Barabasi Lab), Boston, USA
January 2009 – July 2009
- ◇ **Visiting Researcher**, IVIA Ltda., Brazil
September 2005 – 2007
- ◇ **Assistant Professor in Computer Science**, Florida Institute of Technology, USA
August 2000 – 2006
Teaching Activities: Java, Distributed Systems, Network Programming, Algorithms and Data Structures, Coordination Systems, Swarm Intelligence, Introduction to Biologically Inspired Computing.
- ◇ **Lecturer in Computer Science (Assistant Professor)**, University of Ulster, UK
August 1999 – July 2000
Teaching Activities: Object Oriented Programming
- ◇ **Teaching Assistant**, The University of York, UK
August 1996 – July 1999
Teaching Activities: Theory of Computation, Pascal and Java Programming.
- ◇ **Adjunct Faculty in Computer Science**, Padre Anchieta University, Brazil
November 1994 – June 1995
Teaching Activities: Algorithms and Data Structures

INDUSTRIAL

- ◇ **Web Technology Consultant**, Sover8 Limited, UK
February 2000 – July 2000

- ◇ **Systems Consultant**, Mota Machado Constructors, Brazil
January 1993 – June 1995
- ◇ **Computing Services Director**, Fiotex Textile Industry, Brazil
July 1991 – February 1993
- ◇ **Programmer / System Analyst**, Mota Machado Constructors, Brazil
October 1989 – July 1991

TEACHING
ACTIVITIES

GRADUATE COURSES

- ◇ **Complex Networks**
This class is an advanced version of the Social Network Analysis Course. Here students are required to produce a conference quality research paper. The course includes a heavier mathematical formalism on networks and also studies networks that are not social (e.g. technological networks, biological networks, etc.)
- ◇ **Social Networks Analysis**
A seminar-like course where we study social networks using a computational approach. After presentations are made about the basics of social networks the course focuses on how to model social networks, collected data, and analyse the structure of social networks.
- ◇ **Introduction to Biologically Inspired Computing**
A introductory course where students are able to acquire knowledge in most areas of computer science that are inspired by biological systems
- ◇ **Swarm Intelligence**
An advanced course that aimed at senior level graduate students where the area is explored in details. This course was one of the first in the area to be offered worldwide. The material developed has been used by instructors in EPFL (Switzerland) and at the JPL/Caltech (USA).
- ◇ **Coordination Systems**
An advanced course on the theory and practice of coordination. The course provides the students with an opportunity to understand that distributed systems are made of two distinct parts: computation and coordination. This is the first course in the world on the topic.
- ◇ **Distributed Systems**
Standard course on the area given that the field is mainstream computer science.
- ◇ **Network Programming**
A hands-on course where students learn how to write program using concepts they have previously learned on Computer Network courses. Course is heavily based on projects rather than exams.

UNDERGRADUATE COURSES

- ◇ **Fundamentals of Software Development II**
A freshman course on programming – the first for FIT students. The course has been re-organized to encourage students to study hard while providing them with constant feedback on their progress. The course has 4 exams, 10 pop quizzes, 2 semester projects, and 15 lab exercises.
- ◇ **Algorithms and Data Structures**
Arguably the most important course in computer science. The structure is also based on practice in the labs with constant feedback from myself and teaching assistants. This is a standard course in the CS curriculum.

SELECTED INVITED AND KEYNOTE TALKS

- ◇ **Random Thoughts about Causality and Mobility**. European Mobility Symposium. Pisa, Italy. November 2024
- ◇ **Uncovering the differences and similarities between physical and virtual mobility**. Lipari School of Complex Systems. Italy. July 2023
- ◇ **Gender Inequality in Scientific Production**. LANET. Cusco, Peru. August 2023.
- ◇ **Disaggregating Data: A Case of Gender Differences in Collaboration Networks and Urban Human Mobility**. Data Science Africa. Arusha, Tanzania. July 2022.

- ◇ **Differences in the spatial landscape of urban mobility: Gender and socioeconomic perspectives.** Alan Turing Institute. Urban Analytics. Leeds, UK. 2021
- ◇ **Network Science as a Framework for Understanding Population Health.** One Health One Planet Meeting. Lavras, Minas Gerais, Brazil. 2021
- ◇ **Using Data Analytics to Understand Human Mobility Patterns in Cities.** Analytics Forum, Colombia, 2020.
- ◇ **The Next Big Thing in Big Data.** CSoNet. Ho Chi Min. Vietnam, 2019.
- ◇ **On the Structure, Concentration and Dynamics of Urban Crime.** LA-CCI. Guayaquil. Ecuador. 2019
- ◇ **On the Structure, Concentration and Dynamics of Urban Crime.** MARAMI. Dijon. France. 2019
- ◇ **On the Scale, Concentration and Dynamics of Crime.** Urban Analytics Forum. Alan Turing Institute. Bristol. 2019
- ◇ **On the Scale, Concentration and Dynamics of Crime.** Cambridge Networks Day. 2019
- ◇ **On the Scale, Concentration and Dynamics of Crime.** LANET. Cartagena, Colombia. 2019
- ◇ **On the Scale, Concentration and Dynamics of Crime.** Lipari School of Complex Systems. Italy. July 2019.
- ◇ **Human Dynamics and Landscape Decisions: Why People and Urban Systems Matter.** Isaac Newton Institute. University of Cambridge, UK, July 2019.
- ◇ **On the Scale, Concentration and Dynamics of Crime.** University of Aveiro, Portugal. July 2019
- ◇ **On the Scale, Concentration and Dynamics of Crime.** Alan Turing Institute, London. June 2019
- ◇ **Crime Concentration in Cities.** IFISC. Palma, Spain, 2019
- ◇ **Self-Organisation in Computer Science.** Christmas Lecture, University of Exeter. UK. 2018
- ◇ **Spatio-Temporal Regularities in Crime Dynamics.** Oxford Mathematics Institute. UK. 2018
- ◇ **Human Mobility.** LA-CCI, Arequipa, Peru. 2017.
- ◇ **The Scaling of Crime Concentration in Cities.** LANET, Puebla, Mexico. 2017.
- ◇ **Assessing Swarm System using Network Sciences.** NetSci-X 2015, Rio de Janeiro, Brazil. 2015.
- ◇ **Assessing Swarm System using Network Sciences.** NetSci-X 2015, Rio de Janeiro, Brazil. 2015.
- ◇ **Human Mobility: Social Relations Applied to Smart Cities.** Science for the City, Fortaleza, Brazil, October 2013
- ◇ **Using Swarm Intelligence and Network Sciences in Crime Modeling.** Round table at BRICS-CCI, Recife, Brazil, September 2013.
- ◇ **Evaluating the Performance of Social Networks of Sensors under Different Mobility Models.** University of Fortaleza, Brazil, September 2013.
- ◇ **Advances in Network Science fundamental research. How are we doing?** Panel Discussion. IEEE Network Science Workshop. May 2013.
- ◇ **Short Course on Network Science.** AGH University of Science and Technology, Krakow, Poland, June 2012.
- ◇ **Research at BioComplex Lab, Florida Tech.** State University of Pernambuco, May 2011.
- ◇ **Short Course on Complex Networks.** University of Fortaleza, Brazil, August 2010
- ◇ **Short Course on Bio-Inspired Computing.** University of Fortaleza, Brazil, June 2006.
- ◇ **Trends in Information Technology.** Unifor World. University of Fortaleza, Brazil, October 2005.
- ◇ **What Swarms can Teach Computer Scientists.** University of Fortaleza, Brazil, April 2005

STUDENT
SUPERVISION

CURRENT PHD STUDENTS

- ◇ **Guilherme Machado:** Inequality in Social Media
2022-Present.
- ◇ **Andrew Schendl:** Using Mobility to Model Access to Greenspaces
2023-Present.
- ◇ **Julio Taveira:** Using Network Science to Model Interstate Road Access in Brazil
2023-Present.

SELECTED PAST PHD STUDENTS

- ◇ **Liam Berrisford:** Estimating Pollution from Secondary Public Datasets
2020-2024. Currently a Research Software Engineer at the University of Exeter
- ◇ **Sima Farokhnejad:** Modelling of Uncertainties in Cattle Epidemics
2020-2024, PhD. Currently a Post-Doc at Southhampton University, UK
- ◇ **Ana Jaramillo:** Echo Chambers in Science
2019-2023. Currently a post-doc at Technische Universität Graz, Austria
- ◇ **Clodomir Santana:** Using Network Science to Characterise Swarm-based Algorithms
2019-2023. Currently a post-doc at UC Davies, USA.
- ◇ **Josemar F. Cruz:** Social groups characterization and dynamics: a network science approach
2014-2020, Currently a Data Scientist in Canada.
- ◇ **Mahdi Seyedezhad:** A Network Driven Approach for Characterizing Emoji Usage in Social Media
2015-2020, Currently a Senior Data Science working in the Chicago Region, USA.
- ◇ **Diego P. Silva:** A Comprehensive Data-Driven Characterization of Organ Transplantation
2014-2018, Currently Assistant Professor at Universidade Católica de Pernambuco, Brazil
- ◇ **Diogo Pacheco:** Information Densification of Social Constructs via Behavior Analysis of Social Media Users – A Study on Twitter
2013-2017, PhD. Currently a senior lecturer at the University of Exeter, UK
- ◇ **Basim Mahmood:** Social Capital in Wireless Sensor Networks
2012-2015, Currently an Associate Professor at the University of Mosul.
- ◇ **Hugo Barbosa-Filho:** Frequency and Recency of Visits in Human Mobility
2012-2016, Currently a Lecturer in Urban Systems. University of Exeter, UK.

SERVICE ACTIVITIES

STEERING COMMITTEE AND BOARDS

- ◇ **Board Member, Latin American Conference on Complex Networks:** 2017-Present
- ◇ **Co-Editor-in-Chief of Applied Network Science Journal, Springer Verlag:** 2015-Present
- ◇ **Board Member, Complexity Series, Springer Verlag:** 2016-Present
- ◇ **Board Member, Complex Systems Society:** 2023-Present
- ◇ **Co-Chair, Conference on Complex Networks (CompleNet):** 2009-Present
- ◇ **Associate Editor, Complexity Journal,** 2018-Present
- ◇ **Editor. PLoS One,** 2020-Present
- ◇ **Associate Editor, Entropy Journal,** 2024-Present
- ◇ **Associate Editor, Learning & Nonlinear Models:** 2016-2024
- ◇ **Board Member, Network Science Society:** 2013-2025
- ◇ **IEEE Task Force on Network Science:** 2011-2013.
- ◇ **ACM SIGAPP:** Special Interest Group on Applied Computing, 2003-2009
- ◇ **International Journal of Aviation Technology, Engineering and Management,** 2011-2013

RECENT CONFERENCE ORGANISATION

- ◇ **General Co-Chair.** CompleNet 2025. April 2025. Fortaleza, Brazil
- ◇ **Program Co-Chair.** Conference on Complex Systems. September 2024. Exeter, UK

- ◇ **General Co-Chair.** CompleNet 2024. April 2024. Exeter, UK
- ◇ **Organising Committee Member.** NetSci-X. Venice, Italy. 2024
- ◇ **Organising Committee Member.** Conference on Complex Systems (CCS 2023). Salvador, Brazil. October 2023 (to take place).
- ◇ **Co-Chair.** CompleNet 2023. Aveiro, Portugal. April 2023.
- ◇ **Co-Chair** GoodIT. Limassol. Cyprus. 2022
- ◇ **General Chair** CompleNet Live. Exeter. UK. 2020-2022
- ◇ **General Chair** CompleNet. Exeter. UK. 2020
- ◇ **Program Chair** Ideal. Manchester, 2019
- ◇ **Organizing Committee Member** NetSci-X, Warclaw, Poland, 2016.
- ◇ **Chair**, NetSci-X, Rio de Janeiro, 2015.
- ◇ **Chair**, Workshop on Complex Networks (CompleNet), 2009 – Present.

REFERENCES Available upon request